

Predicting the efficacy of malaria vaccines across the life-course

Supplementary Notebook 11. Estimates of distributions of blood infection rates

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1 Introduction

Mathematical models of the epidemiological effects of malaria vaccines demonstrate that vaccine efficacy and cases averted depend on transmission intensity (e.g., Penny et al. (2016)). In addition, our modelling demonstrates that long-term averted deaths and death efficacy are highly sensitive to transmission intensities below ~3 blood infections per year for pre-erythrocytic vaccines that temporarily reduce the risk of infection and may even go negative below ~1.5 blood infections per year. The decision of which vaccines to deploy in a population should therefore be informed by the prevailing blood infection rate. It is of interest, therefore, to quantify the distributions of blood infection rates in children in Africa where *P. falciparum* malaria exists.

Blood infection rate, however, has not been used as a measure of transmission intensity in the malaria epidemiological literature due to the difficulty of measuring independent blood infections, particularly in high transmission settings. Instead, the standard measures of transmission intensity are the entomological inoculation rate (EIR) and the parasite rate ($PfPR_{2-10}$) (Smith et al. 2007). EIR is typically measured as the average number of infectious bites per person per year - which is estimated by multiplying the human biting rate (how often mosquitoes bite people) by the sporozoite rate (the percentage of those mosquitoes carrying infectious malaria parasites). $PfPR_{2-10}$ is the proportion of two-to-ten year olds in a population with asexual *P. falciparum* parasites in their blood at a given point in time. These rates provide broad estimates of transmission intensity but imprecise estimates of average blood infection rates in a population.

The key source of data on worldwide transmission intensity, as measured by $PfPR_{2-10}$, is the Malaria Atlas Project (MAP, Hay and Snow (2006)). MAP has collated $PfPR_{2-10}$ estimates in thousands of sites around the world over the last four decades — the majority in Africa. This survey data is used to infer the geo-spatial distribution of $PfPR_{2-10}$, which can be downloaded from the MAP website.

Our aim here is to first construct distributions of $PfPR_{2-10}$ over two datasets: (i) the MAP survey data in Africa and (ii) the number of African children under ten years of age.

To obtain the distribution of $PfPR_{2-10}$ over survey sites, we downloaded the survey data for *P. falciparum* by cloning the malaria-atlas-project github repository and running the following command in R:

```
getPR(species = "Pf")
```

To obtain the distribution of $PfPR_{2-10}$ over African children under ten we downloaded a raster of the geospatial distribution of $PfPR_{2-10}$ from the MAP website. This map has a spatial resolution of 5km. And we downloaded the 2025 worldwide spatial distribution of people from WorldPop which has a spatial resolution of 1km (Bondarenko et al. 2025).

Having constructed distributions of $PfPR_{2-10}$, we then convert them into distributions of blood infection rates. To achieve this we make use of two datasets that quantify the relationship between $PfPR_{2-10}$ and annual clinical episode rate in the under fives (Griffin, Ferguson, and Ghani 2014; Trape et al. 2014) and combine

these with our model in which we quantify the relationship between annual clinical episode rate in the under fives with blood infection rate.

2 Distribution of $PfPR_{2-10}$ over MAP surveys

Figure 1 shows a the distributions of $PfPR_{2-10}$ every five years from 1987 to 2022 (Hay and Snow 2006). The distribution of $PfPR_{2-10}$ has remained relatively stable over 35 years of survey data. For our analysis we chose the most recent survey at each site from 2020 onwards ($n = 2757$ sites). The distribution of these sites is represented by the thick black line.

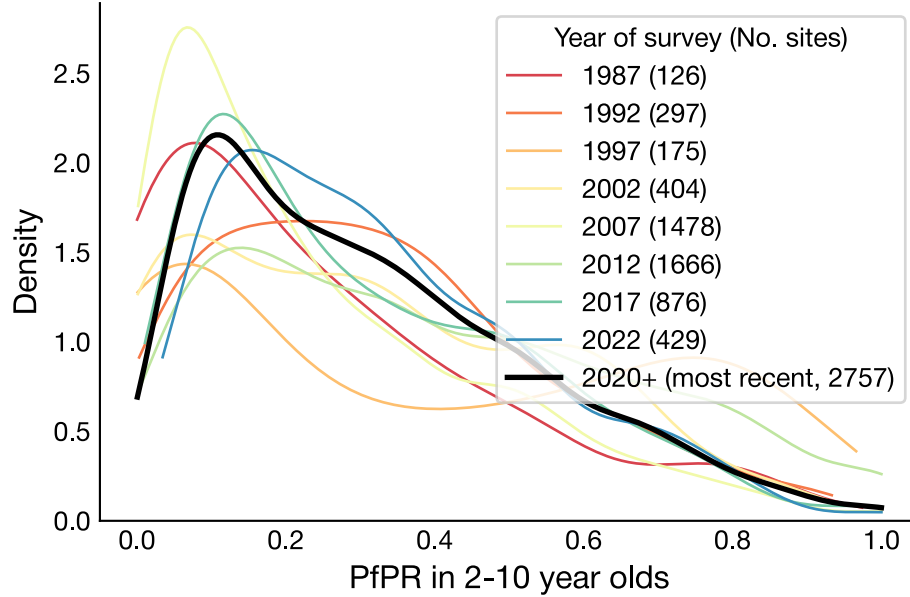


Figure 1: Kernel density estimation plots of $PfPR_{2-10}$ survey data from the Malaria Atlas Project. Coloured lines are data from the indicated single years. The thick black line is the most recent survey at each site from 2020 onwards.

3 Distribution of $PfPR_{2-10}$ over African children under ten

3.1 Geospatial distribution of $PfPR_{2-10}$

The geospatial distribution of $PfPR_{2-10}$ is shown in Figure 2 where higher $PfPR_{2-10}$ is represented by brighter red.

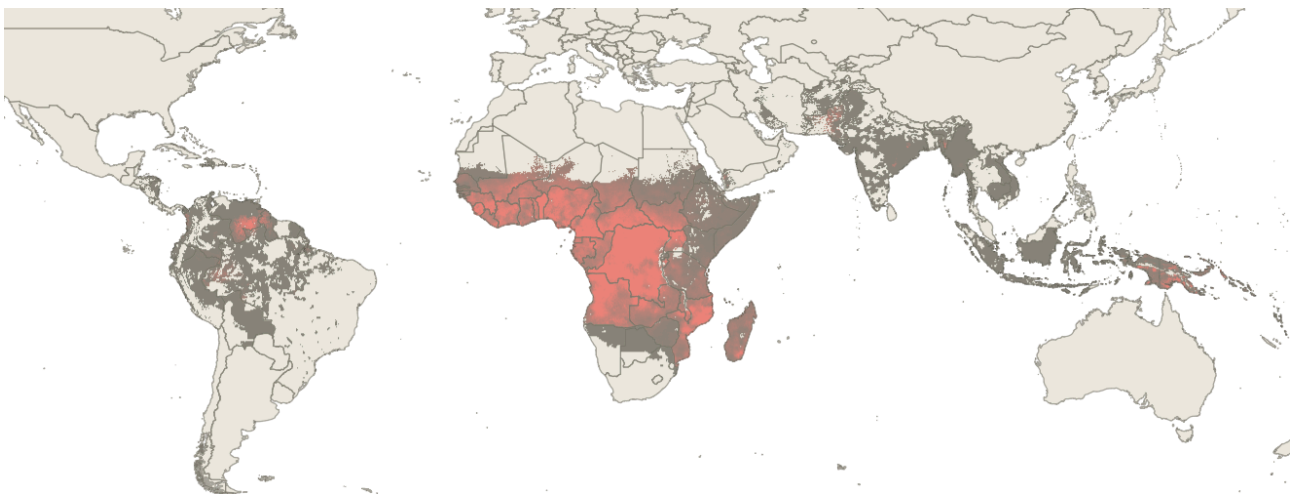


Figure 2: Geospatial distribution of posterior mean estimates of $PfPR_{2-10}$ downloaded from the Malaria Atlas Project. Resolution is 5km. Brighter red denotes higher $PfPR_{2-10}$.

3.2 Geospatial distribution of children under ten

The geospatial distribution of children under ten is shown in Figure 3 as green pixels.

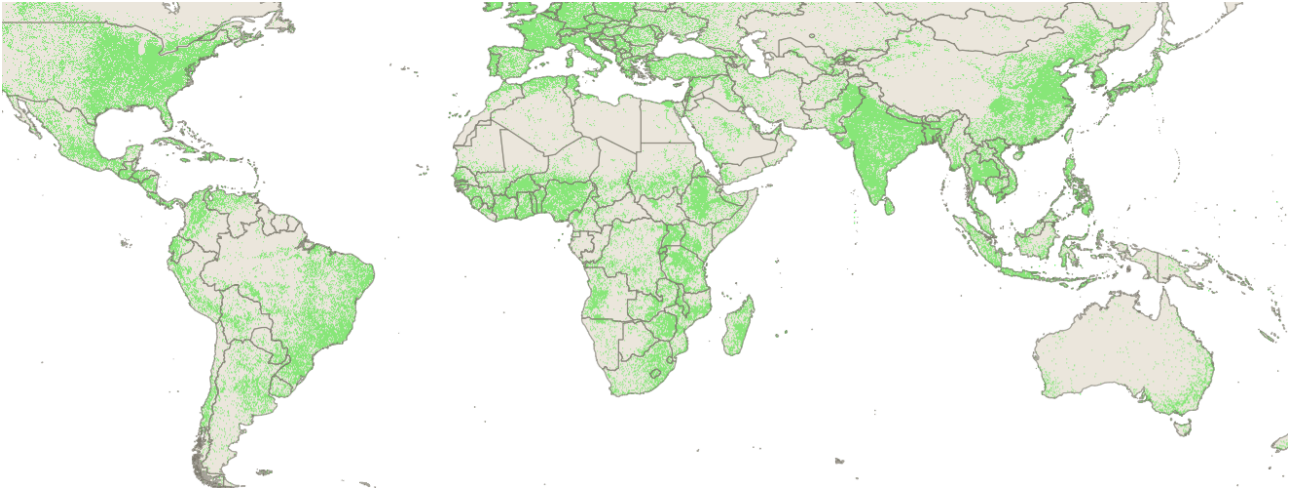


Figure 3: Geospatial distribution of children under ten in 1km² pixels downloaded from WorldPop.

For each 1km² pixel containing at least one child under ten years of age we extracted the corresponding value of $PfPR_{2-10}$ if its value was above 0.1% and we extracted the number of children under 1, between 1-4 years old and between 5-9 years old. The code can be found in `geotiff.ipynb`.

Figure 4A shows the distribution over the whole world (393m children under ten) and over Africa-alone (351m children under ten). The main difference is the peak at very low transmission due to South East Asia and South America. In Africa, there is a sharp peak at about 4% due to low transmission mainly in East and Southern African. Panel B stratifies African children into the under 1s, 1-4 year olds and 5-9 year olds. There is little difference in the shapes of the distributions, but they vary in scale due to the different age ranges.

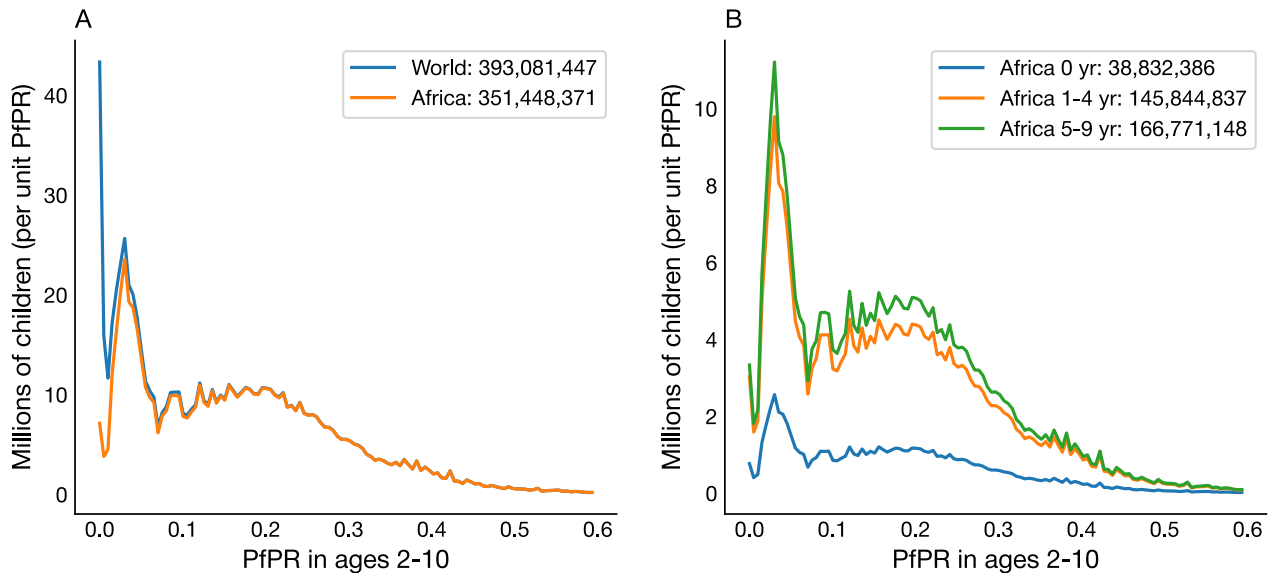


Figure 4: Distributions of $PfPR_{2-10}$. A) whole world and Africa-alone. B) African children stratified by age.

The shapes of the distributions differ between surveys Figure 1 and African children Figure 4. In particular, the survey data peaks at around 10% and linearly decays to 0 at 100% whereas the African children data is bimodal with peaks around 4% and 20% and decays rapidly to zero above 60%. This may indicate a bias in surveys toward being conducted in sites with more prevalent malaria.

4 Distributions of blood infection rate

Having constructed distributions of $PfPR_{2-10}$, we now convert them into distributions of blood infection rates. To achieve this we first analyse two datasets that quantify the relationship between $PfPR_{2-10}$ and annual clinical episode rate in the under fives (Griffin, Ferguson, and Ghani 2014; Trape et al. 2014). We then combine that analysis with simulations from our model in which we quantify the relationship between annual clinical episode rate in the under fives with blood infection rate.

4.1 The relationship between $PfPR_{2-10}$ and clinical episode rate

Patil et al. (2009) have previously modelled the relationship between $PfPR_{2-10}$ and clinical episodes per 1,000 individuals per year. The relationship plateaus above a $PfPR_{2-10}$ of 20% in Africa and above 2% in Asia and the Americas (see Figure 5 of Patil et al. (2009)). This means that this measure of clinical episode rate contains little information about $PfPR_{2-10}$ in the plateau. In turn, this prevents us from quantifying a relationship between $PfPR_{2-10}$ and blood infection rate.

A more information rich measure of clinical episode rate is the average annual clinical episodes in the under fives as these children are still developing clinical immunity so their clinical episode rate is more predictive of blood infection rate and, therefore, $PfPR_{2-10}$ than in the wider population.

We found two datasets in the literature that contain this information: Figure 5A, reproduced from Figure 1 of Griffin, Ferguson, and Ghani (2014), shows the relationship between $PfPR_{2-10}$ and average annual clinical episodes in under fives in twenty three African sites. For the majority of these sites the definition of a clinical episode was fever and detectable parasites by microscopy; although a handful of sites used higher parasitaemia thresholds. Case detection was variable: active or passive, daily or weekly. Five sites did not report $PfPR_{2-10}$; instead values were estimated from EIR (Griffin, Ferguson, and Ghani 2014).

Figure 5A demonstrates that there is no one-to-one relationship between clinical episode rate and $PfPR_{2-10}$. But they are, at least, positively correlated and there is no obvious plateau.

A higher quality dataset, albeit from just one site — Dielmo in Senegal, is shown in Figure 5B (Trape et al. 2014). Clinical episodes were extracted from Figure 4A, and $PfPR_{2-10}$ was extracted from Figure S2A, of Trape et al. (2014). This site has been uninterruptedly followed since 1990 with a strict clinical surveillance program (Trape and Rogier 1996; Trape et al. 2024). This includes daily home visits to each person and the 24-hour presence of medical teams in the village.

The data show a clear positive relationship over a wider range of clinical episode rates than in panel A. This likely reflects the active, frequent and accurate diagnosis of clinical episodes in Dielmo.

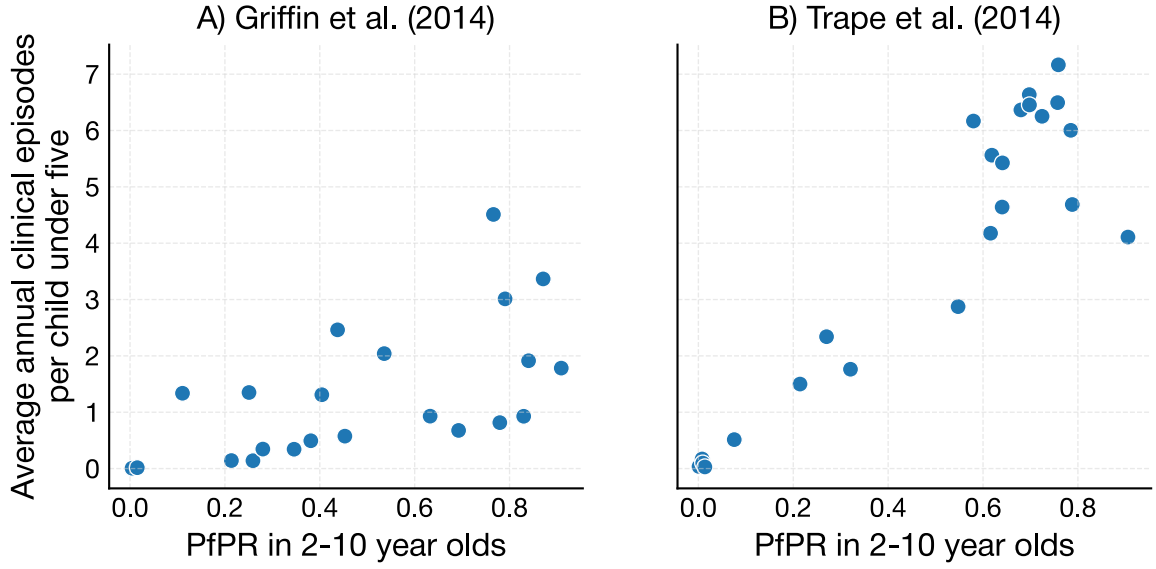


Figure 5: Relationship between $PfPR_{2-10}$ and average annual clinical episodes per child in the under fives in A) 23 African sites (extracted from Figure 1 of Griffin, Ferguson, and Ghani (2014).) and B) Dielmo, Senegal (extracted from Figures 4A and S2A of Trape et al. (2014)).

A linear regression of log-average annual clinical episodes in the under fives against log- $PfPR_{2-10}$ is shown in Figure 6 for both datasets. (Logs were taken to mitigate against heteroscedasticity.) The regression parameters are given in Table 1.

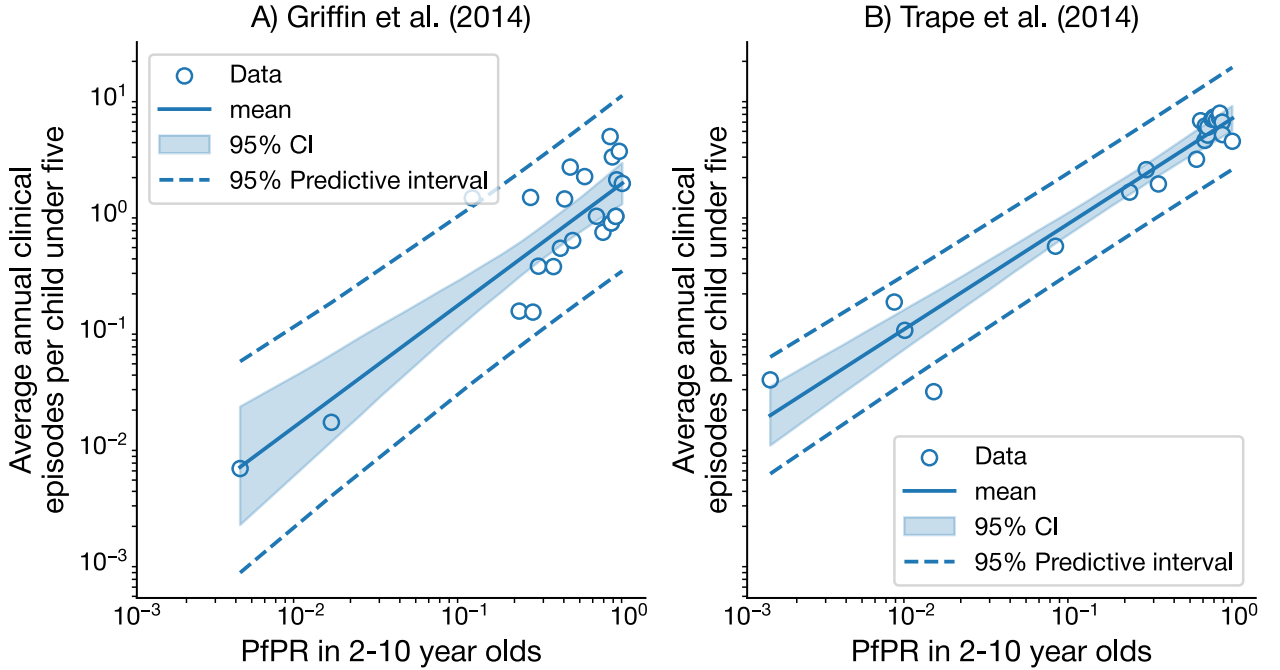


Figure 6: Linear regression of log-average annual clinical episodes in the under fives against $PfPR_{2-10}$. Solid line: mean estimate, dashed lines: 95% predictive interval band estimated from the regression.

Parameter	Griffin, Ferguson, and Ghani (2014)	Trape et al. (2014)
Intercept, β_0	$0.30 \pm 0.09^\dagger$	0.85 ± 0.05
Slope, β_1	1.05 ± 0.13	0.91 ± 0.05
σ_c	0.39	0.23

Table 1: Regression parameters. † Mean $\pm$ standard error

The intercepts differ substantially between the two datasets, but the slopes are similar at about 1, suggesting that, on average, annual clinical episodes in the under fives is proportional to $PfPR_{2-10}$. The data from Dielmo suggest a numerical relationship of the form

$$\text{Mean average annual clinical episodes in the under fives} \approx 7 \times PfPR_{2-10}$$

which is what we should have expected from eye-balling Figure 5B.

The analysis also suggests that variability in average annual clinical episodes increases with $PfPR_{2-10}$. Of key interest is the 95% predictive interval represented by the dashed lines in Figure 6. This represents the width of the distribution of observable average annual clinical episodes in the under fives for a given $PfPR_{2-10}$. We will assume that this distribution is normal with standard deviation given by the average width of the predictive interval across observed $PfPR_{2-10}$ divided by 2×1.96 , which we call σ_c .

Given the higher quality and smaller variability of the Trape et al. (2014) data, we only use its regression parameters in our further analysis.

4.2 The relationship between annual clinical episodes in the under fives and blood infection rate

We can use our epidemiological model to obtain the relationship between blood infection rate and average annual clinical episodes in the under fives. The sensitivity of this relationship to changes in model parameters is shown in Figure 7.

This relationship should be sensitive to parameters that affect the risk of infection and the risk an infection becomes clinical in the first few years of life. This premise is born out in the analysis shown in Figure 7. The relationship is sensitive to the early life protection from infection parameters ρ_{elp} and τ_{elp} and to the risk of clinical malaria on first infection $\rho_{clinical}$.

The relationship is insensitive to changes in bite rate/susceptibility heterogeneity, β and rate of decline in risk of clinical malaria with exposure $\delta_{clinical}$. The latter only affects the risk of clinical malaria later in life.

The relationship is also insensitive to changes in the half-maximum-risk of clinical malaria $\gamma_{clinical}$ below about ten blood infections per year. However, at higher infection rates the relationship is sensitive to this parameter because children reach this threshold earlier in life.

In our analysis below we will consider how this sensitivity to model parameters affects the distribution of blood infection rate over African children.

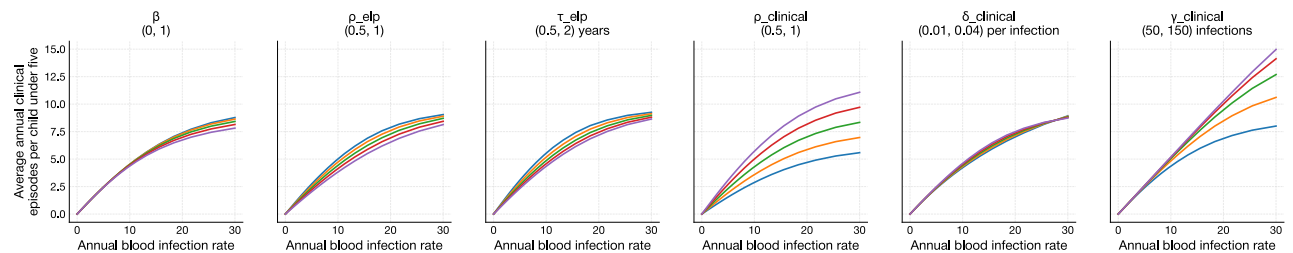


Figure 7: Sensitivity of the relationship between average annual clinical episodes in the under fives and blood infection rate to changes in epidemiological model parameters.

4.3 Distributions of average annual clinical episodes in the under fives and blood infection rate

Here we detail how we convert the distributions of $PfPR_{2-10}$ — either over survey sites as shown in Figure 1 (black line) or over African children under ten as shown in Figure 4 (orange line) — to distributions of blood infection rate via distributions of average annual clinical episodes in the under fives.

First we propagate the uncertainties in the estimates of β_0 and β_1 and the variation in the $PfPR_{2-10}$ -average annual clinical episodes in the under fives relationship, σ_c , into the distribution of average annual clinical episodes in the under fives. In the following all logs are in base 10.

Let $p(\log c \mid \log r, \beta_0, \beta_1)$ be the distribution of log-average annual clinical episodes in the under fives ($\log c$) for a given log- $PfPR_{2-10}$ ($\log r$), β_0 and β_1 (Figure 6). Let $p(\beta_0, \beta_1)$ be the joint distribution of β_0 and β_1 representing the uncertainty in their estimates. We assume that $p(\log c \mid \log r, \beta_0, \beta_1)$ has a normal distribution with mean $\beta_0 + \beta_1 \log r$ and variance σ_c^2 . We assume $p(\beta_0, \beta_1)$ has a multivariate normal distribution with mean $(\beta_0 \ \beta_1)$ and co-variance matrix $\begin{pmatrix} \sigma_{\beta_0}^2 & \sigma_{\beta_0\beta_1}^2 \\ \sigma_{\beta_0\beta_1}^2 & \sigma_{\beta_1}^2 \end{pmatrix}$, which we obtain from the linear regression of the Trape et al. (2014) data above. Then, the distribution of log-average annual clinical episodes in the under fives given a log- $PfPR_{2-10}$ is

$$\begin{aligned} p(\log c \mid \log r) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(\log c \mid \log r, \beta_0, \beta_1) p(\beta_0, \beta_1) d\beta_0 d\beta_1 \\ &= \mathcal{N}(\beta_0 + \beta_1 \log r, \sigma_c^2 + \sigma_{\beta_0}^2 + 2\sigma_{\beta_0\beta_1}^2 \log r + \sigma_{\beta_1}^2 [\log r]^2) \end{aligned}$$

where $\mathcal{N}(\cdot)$ represents a normal distribution.

Next, let $p(\log r)$ be a distribution of $PfPR_{2-10}$ either over survey sites or African children under ten. Then the distribution of log-average annual clinical episodes in the under fives is

$$p(\log c) = \int_0^1 p(\log c \mid \log r) p(\log r) d \log r$$

This integral is solved numerically (Scripts/infection_rate.py). To transform from log-average annual clinical episodes in the under fives to average annual clinical episodes in the under fives we use the change of variable formula to obtain

$$\begin{aligned} p(c) &= p(\log c) \frac{d \log c}{dc} \\ &= p(\log c) \frac{10^{-\log c}}{\ln 10} \end{aligned}$$

Finally, to transform from average annual clinical episodes in the under fives to blood infection rate we use a particular relationship from Figure 7 with the change of variable formula:

$$p(\lambda) = p(c) \frac{dc}{d\lambda}$$

where we solve $dc/d\lambda$ numerically.

The distributions of $PfPR_{2-10}$ over survey sites (blue line) and over African children under ten (orange line) are shown in Figure 8A. The cumulative distributions are shown in Figure 8D.

The inferred distributions of average annual clinical episodes in the under fives are shown in Figure 8B, E. The inferred distributions of blood infection rates are shown in Figure 8C, F using the model default parameters which are estimated from the population-level clinical data from Dielmo and Ndiop villages, Senegal (Notebook S5, Trape and Rogier (1996), Trape et al. (2024)).

Although the distributions of $PfPR_{2-10}$ over survey sites and over African children under ten differ markedly, there is less difference in the distributions of average annual clinical episodes in the under fives and blood infection rate. This is because there is considerable variability in the relationship between $PfPR_{2-10}$ and clinical episode rate (Figure 6).

In conclusion, the distribution of blood infection rate in African children is insensitive to uncertainty in, and to some extent differences between, the distributions of $PfPR_{2-10}$.

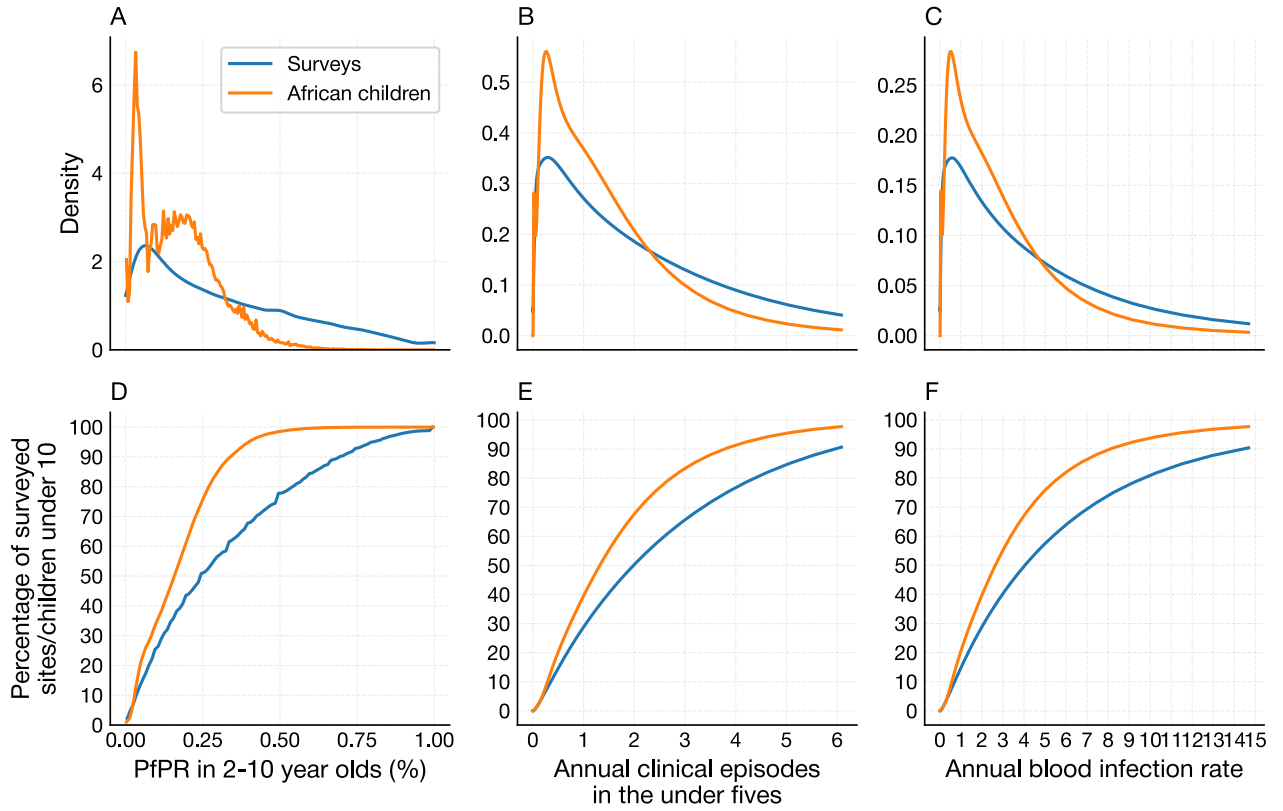


Figure 8: A) Distributions of $PfPR_{2-10}$ over survey sites (blue) and over African children under ten (orange). B) Inferred distributions of average annual clinical episodes in the under fives. C) Inferred distributions of blood infection rates. D-F) cumulative distributions of A-C. Early life protection and clinical immunity parameters are the posterior means of the homogeneous model fitted to the population-level data of Dielmo and Ndiop village, Senegal.

5 Sensitivity of blood infection rate distribution over African children under ten to changes in parameter values

Figure 9 demonstrates the sensitivity of the distribution of blood infection rate distribution over African children under ten to changes in parameters ρ_{elp} , τ_{elp} and $\rho_{clinical}$.

Changes in these parameters do not change the position of the distribution's peak, but they do broaden the distribution (top row of Figure 9). This changes how rapidly the cumulative distribution rises with increasing infection rate (bottom row). So, given a particular infection rate there is some uncertainty in how many African children under ten experience at most this infection rate. The degree of uncertainty depends on infection rate. But, overall, we can conclude that quantifying the distribution of blood infection rate across Africa is possible with reasonable precision.

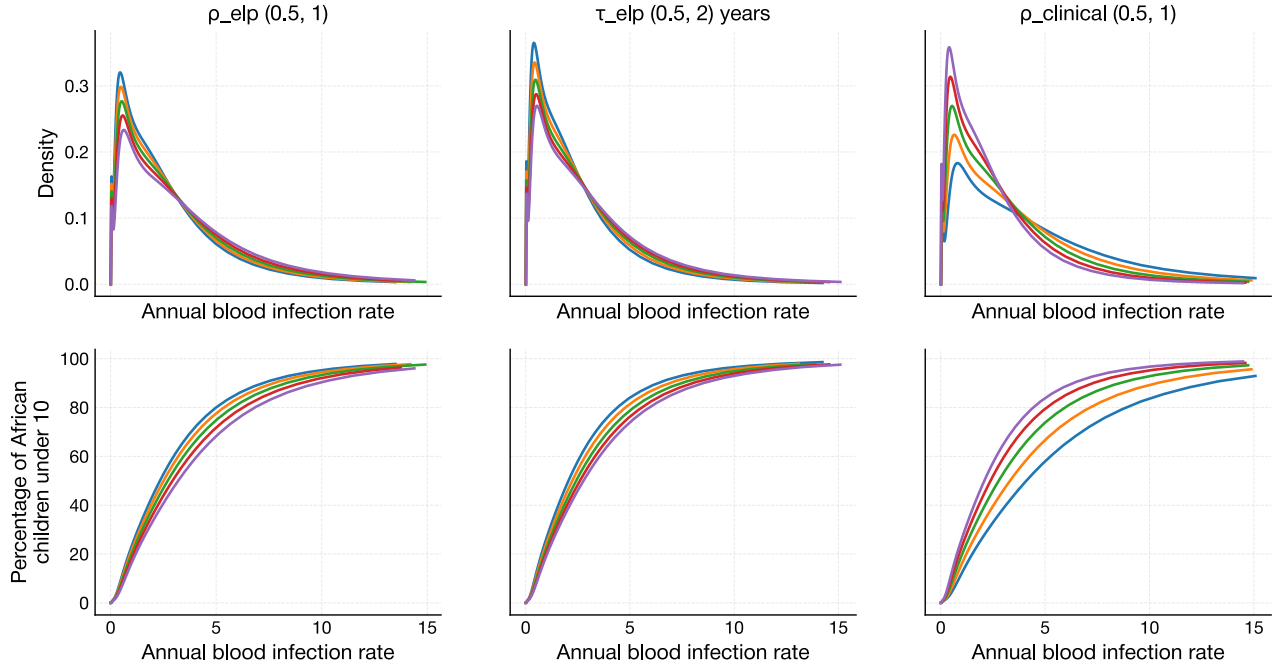


Figure 9: Sensitivity of A) distributions and B) cumulative distributions of annual blood infection rate over African children under ten to changes in relative protection of newborns to infection, ρ_{elp} , half-life of early life protection, τ_{elp} , and the risk of clinical malaria on first infection, $\rho_{clinical}$.

6 R21 deaths averted weighted by distribution of blood infection rate

Having quantified the distribution of infection rates across Africa, we can then ask, if all African children were vaccinated with R21 with one booster, how many deaths would be averted?

Assuming our default epidemiological model parameter values, we have plotted the lifetime number of deaths averted per 100,000 fully R21-vaccinated children with one booster in Figure 10B. Given these parameter values, there exists a critical blood infection rate (λ_{R21}) of about 1.5 blood infections per year, below which averted deaths is negative. This is because R21 temporarily slows infection rate, slowing the development of natural immunity causing more severe episodes in the over fives which have a higher mortality rate than at younger ages.

In Figure 10C we calculate averted deaths weighted by the distribution of blood infection rate across Africa. The answer to the above question “If all African children were vaccinated with R21, how many deaths would be averted?” is the area under the curve in panel C.

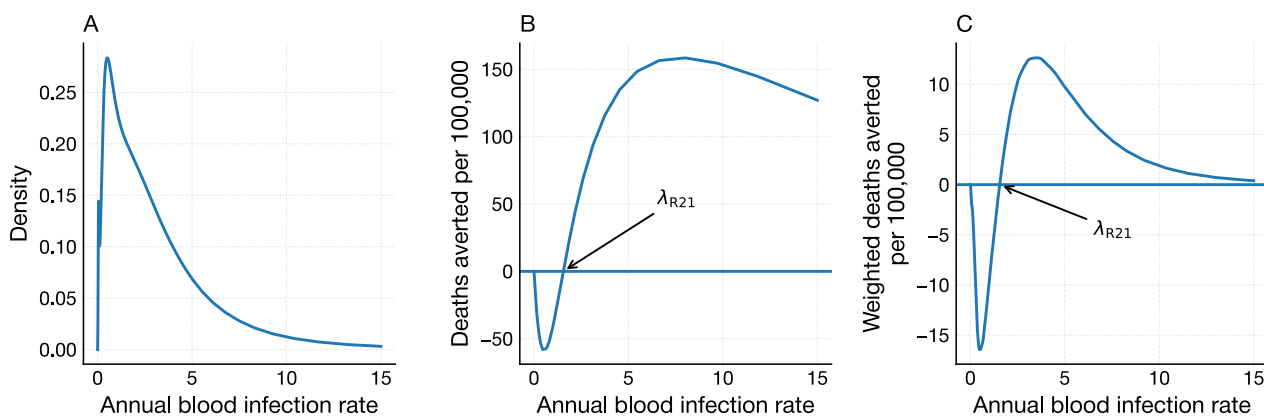


Figure 10: A) Distribution of annual blood infection rate over African children under ten. B) Lifetime deaths averted per 100,000 fully R21-vaccinated children with one booster in a seasonal setting across blood infection rates for the default model parameters. C) Lifetime deaths averted weighted by the distribution of blood infection rate.

The area under the curve can be decomposed into deaths indirectly caused by R21 and deaths directly averted by R21. In addition, we can also calculate the ratio of deaths averted to deaths caused by taking the ratio of the areas under the curves to the right and left of λ_{R21} respectively.

These values are shown in Figure 11 for different values of ρ_{elp} , τ_{elp} , $\rho_{clinical}$ and ϵ_{severe} — the parameters that have the most effect on the distribution of blood infection rate and λ_{R21} (see Notebook S10).

We predict that immunising all African children with R21 with one booster i) in sites with blood infection rates below λ_{R21} would indirectly cause about 10-20 deaths per 100,000 fully vaccinated children (top row), ii) in sites with blood infection rates above λ_{R21} would avert about 70-80 deaths per 100,000 fully vaccinated children (second row). The net averted deaths is around 60-70 per 100,000 fully vaccinated children (third row). For every death indirectly caused by R21, R21 will avert between about 4-10 deaths depending on particular parameter values.

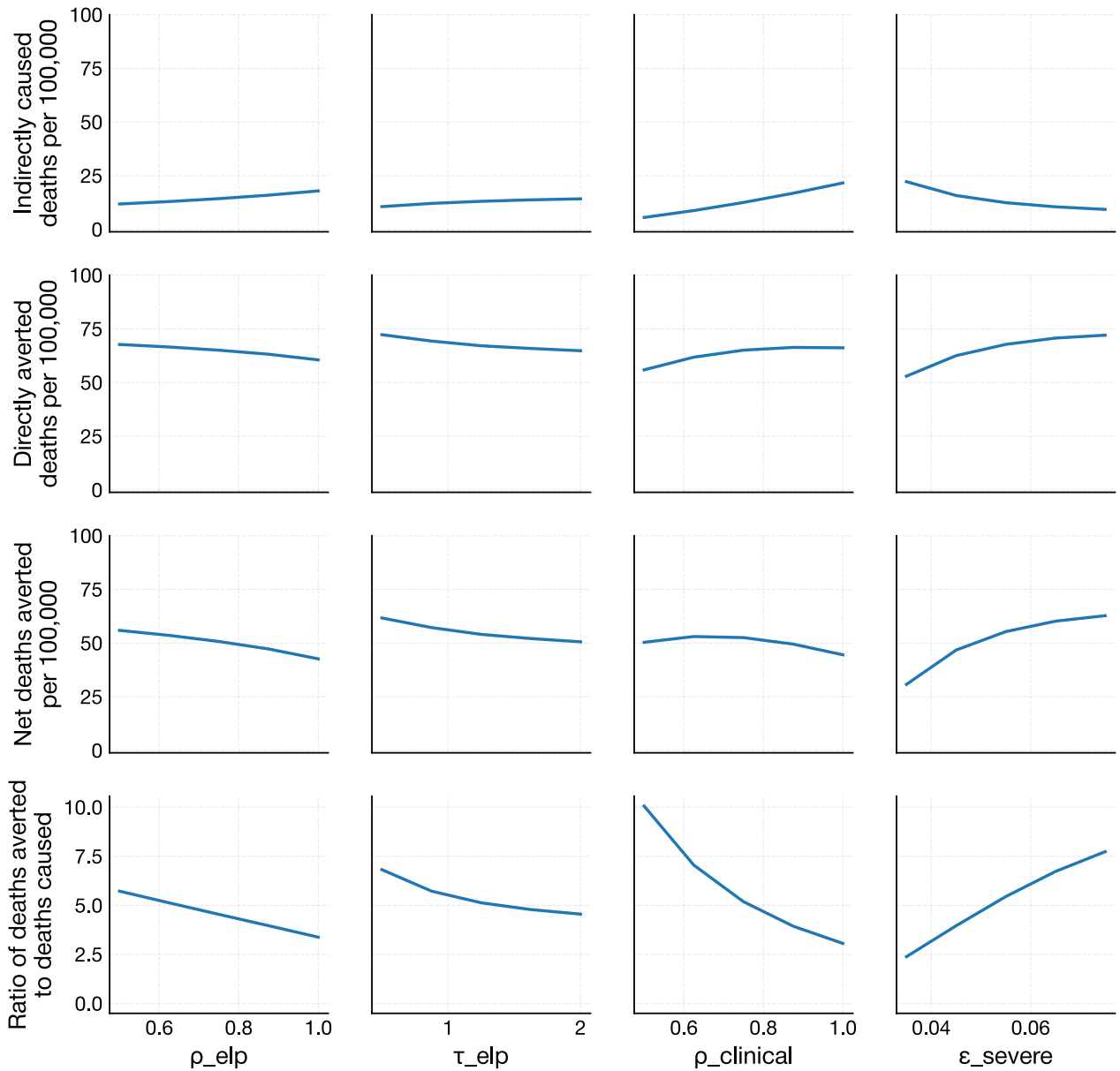


Figure 11: Sensitivity of net averted deaths in 100,000 fully R21-vaccinated African children with one booster in a seasonal setting to changes in parameter values that have the most effect on the distribution of blood infection rate and λ_{R21} . Top row: Indirectly caused deaths per 100,000 (the area under the curve to the left of λ_{R21} in Figure 10C). Second row: Directly averted deaths (the area under the curve to the right of λ_{R21}). Third row: Net averted deaths. Fourth row: the ratio of directly averted deaths to indirectly caused deaths.

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